Theorem 1.1. (The Division Theorem for $\mathbb{Z}$): $a, b \in \mathbb{Z}$ with $b > 0$ $\exists!\, q, r \in \mathbb{Z}$ with $0 \leq r < b$ s.t. $a = bq + r$.

Proof. (*Existance*): Let $R = \{a - bm : m \in \mathbb{Z}\ \&\ a - bm \geq 0\}$, then as: if $a \geq 0$, $a - b \cdot 0 = a \geq 0$ so $a \in R$; if $a < 0$, $0 = a - b \cdot a = a(1-b) \geq 0$ so $a - b \cdot a \in R$, $R \neq \varnothing$. Thus, take $r = \min(R)$ and so $0 < r = a - bm \in R$ for some $m \in \mathbb{Z}$, choose $q = m \in \mathbb{Z}$. So by construction $a = bq + r$. Note, $r < b$ as were such not true then $r > a - b(q+1) = r - b \in R$ but this contradicts $r = \min(R)$, so we have $0 \geq r < b$. (*Uniqueness*): Suppose $q, q', r, r' \in \mathbb{Z}$ with $0 \leq r', r < b$ and $a = bq + r = bq' + r'$, also WLOG take $r' \leq r$. So with $\frac{r-r'}{b} \in (-1, 1)$, and $\frac{r-r'}{b} = \frac{a-bq-(a-bq')}{b} = q' - q \in \mathbb{Z}$, we have $\frac{r-r'}{b} = 0$ i.e. $r = r'$. So $q = \frac{a-r}{b} = \frac{a-r'}{b} = q'$. □

Euclidian Algorithm:
Let $a, b \in \mathbb{Z}^+$, apply Th^m 1.1, to give:

$$\begin{aligned} a &= bq_1 + r_1 \\ b &= r_1 q_2 + r_2 \\ r_1 &= r_2 q_3 + r_3 \\ &\vdots \\ r_{n-1} &= r_n q_{n+1}, \end{aligned}$$

Here, $\gcd(a, b) = r_n$. To prove this relation $G(r_i, r_{i+1}) = G(r_{i+1}, r_{i+2})$, where $G(a, b) = \{n \in \mathbb{Z} : n|a\ \&\ n|b\}$.
Reverse Euclid:
Given a completed algorithm we can obtain $n, m \in \mathbb{Z}$ s.t. $na + mb = \gcd(a, b)$ i.e. taking the end of a Euclidian algroithm:

$$\begin{aligned} &\vdots && (1) \\ r_{n-3} &= r_{n-2} q_{n-1} + r_{n-1} && (2) \\ r_{n-2} &= r_{n-1} q_n + r_n && (3) \\ r_{n-1} &= r_n q_{n+1} \end{aligned}$$

we have:

$$\begin{aligned} (3) &\implies r_n = r_{n-2} - r_{n-1} q_n && (4) \\ (2) &\implies r_{n-1} = r_{n-3} - r_{n-2} q_{n-1} && (5) \\ (4) \text{ and } (5) &\implies r_n = r_{n-2} - \\ &\qquad (r_{n-3} - r_{n-2} q_{n-1}) q_{n-1} \\ &\vdots \end{aligned}$$

we continue this ladder up, simplifying each time, (Note: $r_n = \gcd(a, b)$)
Theorem 1.9. Every integer $n \geq 2$ can be written as a product of primes.

Proof. (*By strong induction*):
(*Base case*, $n = 2$): Trivially true, $n = 2$ a product of times.
(*Inductive Hypothesis*): Suppose for $n \in 1, 2, \ldots, k$ $n = p_{n,1} p_{n,2} \cdots p_{n,m_n}$.
Then for $n = k+1$, if $k+1$ is prime we are done, if not then $k+1 = ab$ where $a, b \in \{2, 3, \ldots, k\}$, thus $a = p_{a,1} p_{a,2} \cdots p_{a,m_a}$ and $b = p_{b,1} p_{b,2} \cdots p_{b,m_b}$ so: $k+1 = p_{a,1} p_{a,2} \cdots p_{a,m_a} p_{b,1} p_{b,2} \cdots p_{b,m_b}$.
So $\forall 2 < n \in \mathbb{N}$ we are done. □

Theorem 1.10. (Euclid's property of primes) $\forall p \in$ Primes & $\forall a, b \in \mathbb{Z}$. $p|ab \implies p|a$ or $p|b$.

Proof. (*Case* $p|a$): We are done.
(*Case* $p \nmid a$): Then using the Euclidian algorithm we obtain: $1 = na - mp$ with $n, m \in \mathbb{Z}$. So $p|b \iff p|(na - mp)b \iff p|n(ab) - mbp \Longleftarrow p|ab$. □

the chain of implications uses a proposition in the notes which states: $p|a\ \&\ p|b \implies p|a + b\ \&\ p|ab$.
Theorem 1.12. (Fundamental Theorem of Arithmetic) $\forall 2 < n \in \mathbb{N}, \exists!\, p_1, p_2, \ldots, p_r \in$ Primes up to ordering, s.t.: $n = p_1 p_2 \cdots p_r$.

Proof. The existance is from Th^m 1.9. For uniqueness, suppose $p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$, then cancel common primes, suppose the result is not 1 i.e. we are left with $p_1 p_2 \cdots p_n = q_1 q_2 \cdots q_m$ with $p_i \neq q_j, \forall i \in \{1, \ldots, n\}$ and $\forall j \in \{1, \ldots, m\}$, after reindexing. Then $p_1 | q_1 q_2 \cdots q_m$ so $\exists j \in \{1, \ldots, m\} p_1 | q_j$, but q_j is prime so $p_1 = q_j$, $\lightning$. □

Lemma 1.16. $\forall a, b \in R$: $0a = a0 = 0$; $a(-b) = (-a)b = -(ab)$; & $(-a)(-b) = ab$

Proof. $0a + 0 = 0a = (0+0)a = 0a + 0a$ so $0a = 0$. $ab + a(-b) = a(b - b) = a0 = 0 = (a - a)b = ab + (-a)b$ so by the Groups and Geo^m we have $-(ab)$ is the unique element s.t. $ab - (ab) = 0$, $a(-b) = -(ab) = (-a)b$.
$(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. □

Lemma 1.18. (Subring Test) Let R be a ring and $S \subseteq R$. Then S is a subring of R iff the following hold, $\forall r, s \in S$:
1. $1_R \in S$,
2. $r + s, rs \in S$, and
3. $-r \in S$.

Proof. By 1 and 3 we have $-1_R \in S$ so $0_R = 1_R - 1_R \in S$ by 2. So $(S, +)$ is a group by the subgroup test with $(R, +)$. Commutativity of $+$, associativity of $+$ and $\times$, and distributivity of $\times$ over $+$ are all inheritited from R. Note, as they shair operations they shair the identities of the operations, so as, $0_R \neq 1_R$ in R we have $0_S = 0_R \neq 1_R = 1_S$ in S, so (R1)-(R4) are satisfied. □

Lemma 2.2. Suppose $Char(R) = n > 0$, then: (i) $n \cdot r = 0, \forall r \in R$; & (ii) if $m \in \mathbb{Z}^+$ then $m \cdot 1 = 0$ iff $n|m$.

Proof. (i):

$$\begin{aligned} n \cdot r &= \underbrace{r + r + \cdots + r}_{n \text{ times}} \\ &= r(\underbrace{1 + 1 + \cdots + 1}_{n \text{ times}}) = r0 = 0 \end{aligned}$$

(ii): ($\implies$): Let $m \cdot 1 = 0$ with $m \in \mathbb{Z}^+$, applying the division Th$_m$ with $m = nq + r$ where $q, r \in \mathbb{Z}$ and $0 \leq r < n$ so $0 = m \cdot 1 = (nq + r) \cdot 1 = q(n \cdot 1) + r \cdot 1 = q \cdot 0 + r \cdot 1 = r \cdot 1$, so $r = 0$ as otherwise $n > r \in \mathbb{Z}^+$ and $r \cdot 1 = 0$, a $\lightning$ of n's def^n. (*impliedby*): Let $n|m$ i.e. $\exists q \in \mathbb{Z}$ s.t. $m = qn$, so $m \cdot 1 = qn \cdot 1 = q(n \cdot 1) = q \cdot 0 = 0$. □

Lemma 2.5. If S is a subring of R and R is a domain then S is a domain.

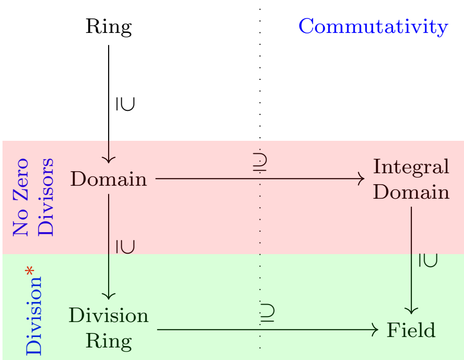
Proof. Let $r, s \in S$ with $rs = 0$, thus $rs = 0$ in R, hence $r = 0$ or $s = 0$. □

Proposition 2.7. Suppose R is a domain. Then $R[X]$ is a domain.

Proof. We use the contrapositive of the definition of a domain [i.e.:R a domain iff $\forall 0 \neq r, s \in R, rs \neq 0$]. Let $0 \neq f, g \in R[X]$, taking the leading coefficients of f & g as $a \neq 0$ and $b \neq 0$, respectively, we the leading coefficent of fg as $ab \neq 0$, hence $fg \neq 0$. □

Note: This immediately (by induction on n) yields $R[X_1, \ldots, X_n]$ and domain if R is a domain.
Lemma 2.11. If R is a ring and $r \in R$ has both a left and right inverse then these equal and unique.

Proof. (Equal): Choose, $rs = 1 = tr$, then: $s = 1s = trs = t1 = t$.
(Unique right inverse): Let $rs = 1 = rs'$ then $s = 1s = trs = t1 = trs' = 1s' = s'$, where t is choosen s.t. $tr = 0$.
(Unique left inverse): Let $tr = t'r$ then $t = t1 = trs = 1s = t'rs = t'1 = t'$, where s is choosen s.t. $rs = 0$ □



*of non-zero elements

Definition 1.4. $\forall a, b \in \mathbb{Z}$, $\gcd(a,b) = \max\{n \in \mathbb{Z} : n|a \ \& \ n|b\}$, noting $\gcd(0,0) = 0$ and if $\gcd(a,b) = 1$ then a and b are coprime.
Definition 1.6. $\forall a, b, c \in \mathbb{Z}$ with $n \geq 2$. a is **congruent to b modulo n** iff $n|a-b$, written $a \equiv b \mod n$.Then define the congruence class of a w.r.t. n as: $[a]_n = \{b \in \mathbb{Z} b \equiv a \mod n\} = \{b \in \mathbb{Z} : b|m-a\}$
We thus have:

$$\mathbb{Z}_n = \{[a]_n : a \in \mathbb{Z}\} = \{[0]_n, [1]_n, [2]_n, \ldots, [n-1]_n\}$$

Note: $b \equiv a \mod n$ iff $b = a + kn$ iff $[a]_n = \{a + nk : k \in \mathbb{Z}\}$ iff $[a]_n = [a+kn]_n$
Definition 1.14. A **ring** is a set R with two binary operations $+$ & $\times$, defined on R, satisfying the following:
(R1) $(R, +)$ is an abelian group with identity 0.
(R2) $\times$ is associative.
(R3) $\times$ distributes over $+$ i.e.:

$$a(b+c) = ab + ac, \ \& \ (a+b)c = ac + bc, \forall a, b, c \in R$$

(R4) The multiplicative identity $1 \in R$ and $1 \neq 0$
Example. $M_n(R)$ is the ring of $n \times n$ matrices, a non-commutative ring. Note: the only ideals of $M_n(R)$ are $M_n(R)$ and $\{\mathbf{0}\}$ but this isn't a division ring hence even relaxing commutative in Pro$^{\text{p}}$ 3.22 doesn't yeild the same pro$^{\text{p}}$ with field $\to$ division ring.
Example. For X a set $\mathcal{P}(X)$ with

$$A + B = (A \cup B) \backslash (A \cap B), \qquad AB = A \cap B$$

and $0 = \varnothing$, $-A = A$, $A, B \in \mathcal{P}(X)$.
Definition 1.17. Let R be a ring and $S \subseteq R$. Then S is a **subring** of R if S is a ring w.r.t. the same addition and multiplication as R and contains the multiplicative identity of R, 1_R.
Definition 1.21. let R be a ring. The **ring of polynomials** $R[X] = \{\sum_{i\geq 0}^{n} a_i X^i \text{a finite sum} : a_i \in R$ with addition and multiplication defined as:
$\sum_{i\geq 0}^{n} a_i X^i + \sum_{i\geq 0}^{m} b_i X^i = \sum_{i\geq 0}^{n} (a_i + b_i) X^i$
&
$(\sum\limits_{i\geq 0}^{n} a_i X^i)(\sum\limits_{i\geq 0}^{m} b_i X^i) = \sum\limits_{i\geq 0}^{n+m} (\sum\limits_{j=0}^{i} a_j b_{i-j}) X^i$
$0 = \sum_{i\geq 0} 0X^i$ & $1 = 1X^0 + \sum_{i\geq 1} 0X^i$. Then the **degree** of a polynomial is $i \in \mathbb{N} \cup \{0\}$ s.t. $a_i \neq 0$ and $\deg(0) = -\infty$
More genearlly, we have:

$$R[X_1, \ldots, X_n] = R[X_1, \ldots, X_{n-1}][X_n]$$

$\forall f, g \in R[X]$, $\deg(f+g) \leq \max(\deg(f), \deg(g))$ and $\deg(fg) \leq \deg(f) + \deg(g)$.
Definition 1.24. If R_1 and R_2 are rings then the **Cartesian product** $R_1 \times R_2$ with the $+$ and $\times$ defined element wise is a ring.
Definition 2.1. For $a \in R$ define: for $n \in \mathbb{Z}^+$, $n \cdot a = \underbrace{a + a + \cdots + a}_{n \text{ times}}$; for $n = 0$, $n \cdot a = 0 \cdot a = 0$; & for $n \in \mathbb{Z}^-$, $n \cdot a = (-n) \cdot (-a)$.
The **Characteristic**, $\text{char}(R)$, of a ring R is the least positive integer n s.t. $n \cdot 1 = 0$.
Note: $(n+m) \cdot a = n \cdot a + m \cdot a$, $n \cdot (a+b) = n \cdot a + n \cdot b$, & $n \cdot (m \cdot a) = (nm) \cdot a$, all easily verifiable.
Definition 2.3. A non-zero element $r \in R$ is a **zero-divisor** if $\exists s \in R$ a non-zero element s.t. $rs = 0$ or $sr = 0$.
A ring R is a domain if, $\forall r, s \in R$:

$$rs = 0 \implies r = 0 \text{ or } s = 0.$$

i.e. a ring with no zero divisors is a domain and vice versa.
Definition 2.9. A **division ring** is a ring where every non-zero element has a right and left inverse, i.e.: $\forall r \in R, \exists s, t \in R$ s.t. $rs = 1 = tr$ with s,t being the right and left inverses, respectively.
Note: this immediately collapse, due to Le$^{\text{m}}$ 2.11, to just there exists an inverse as $s = t$.
An **invertible** element r is called a **unit**.